higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 3

(6050433)

4 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Fillers changes the properties of the surface tension of a liquid.

1.2 Virgin fibre is obtained directly from trees.

1.3 Insecticides kill pets through contact or systematic means.

1.4 Sodium carbonate is used to preserve meat and other substances.

1.5 PPC was founded in 1899 as the first manufacturing plant in South Africa.

(5 × 1) **[5]**

QUESTION 2

Give ONE word/term for each of the following descriptions. Write only the word/term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

2.1 An example of an organic fertiliser that is rich in phosphorus content

2.2 Salty water that is often used to preserve food

2.3 They are used to improve the visual properties of the object that is being cleaned

2.4 A soft white or grey mineral composed of hydrated calcium sulphate

2.5 A mixture of water from wood chips, condensed steam, black liquor and white liquor

(5 × 1) **[5]**

QUESTION 3

3.1 Name FOUR advantages of liquid fertilisers. (4)

3.2 Name THREE secondary nutrients. (3)

3.3 Name FOUR uses of urea which is rich in nitrogen content. (4)

3.4 Draw a labelled process flow diagram for the production of sulphuric acid using pyrite. (13)

[24]

QUESTION 4

4.1 Which calcium compound is used for the following?

4.1.1 Commonly used as a bleaching agent

4.1.2 As flocculent in the water treatment industry

4.1.3 Used by the pulp and paper industry to improve printing properties (3 × 1) (3)

4.2 Give FIVE uses of sodium chloride. (5)

4.3 Name THREE sources of brine. (3)

4.4 Complete and balance the following reaction equation:

$NH_4C\ell(aq) + Ca(OH)_2(aq) \rightarrow$ (6)

4.5 Name THREE properties of cement that makes it versatile. (3)

4.6 Briefly describe how coal or ore is mined by using strip mining. (4)

4.7 Name FIVE stages in sequence used in the slagment production process. (5)

4.8 What are the TWO sources of sodium chloride? (2)

[31]

QUESTION 5

5.1 Soap and detergents are used as cleaning products.

5.1.1 Differentiate between soap and detergents. (2)

5.1.2 Give TWO examples of detergents. (2)

5.1.3 List SIX steps in sequence that is involved during the production of soap through the continuous process. (6)

5.2 Write brief, clarifying notes on fillers. (2)

5.3 Briefly explain boric acid and how it is used to control pests. (3)

5.4 Name TWO primary benefits of using pesticides. (2)

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QUESTION 6

6.1 Define *bagasse.* (2)

6.2 Chip storage is important for the continuous supply of chips in the pulp mill.

6.2.1 Name TWO places where chips can be stored.

6.2.2 What are the TWO negative effects of chips that are stored for a long period?

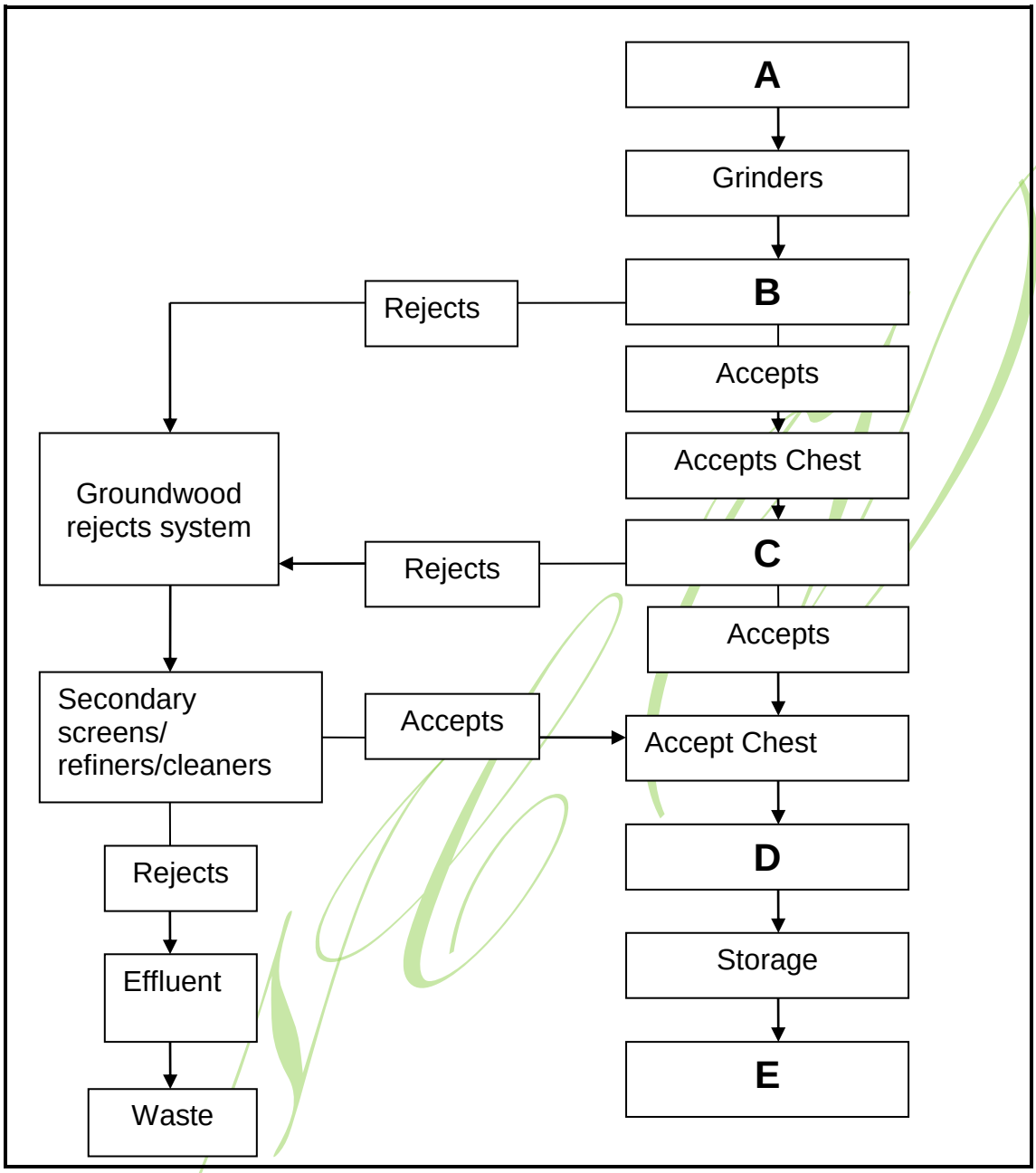
(2 × 2) (4)

6.3 Name THREE phases of the Fourdrinier table. (4)

6.4 A finishing process is performed to improve the paper quality by changing the surface characteristics of the sheet.

Briefly explain the paper finishing operation under the coating process. (3)

6.5 Complete the following diagram of groundwood production by naming the processes (A–E) in the ANSWER BOOK.



(5 × 1) (5)

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TOTAL: 100

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